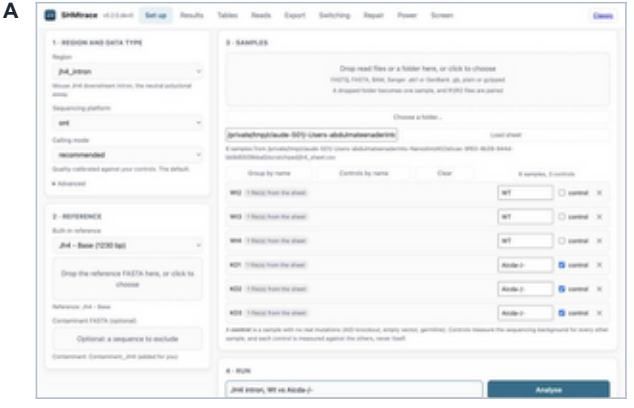
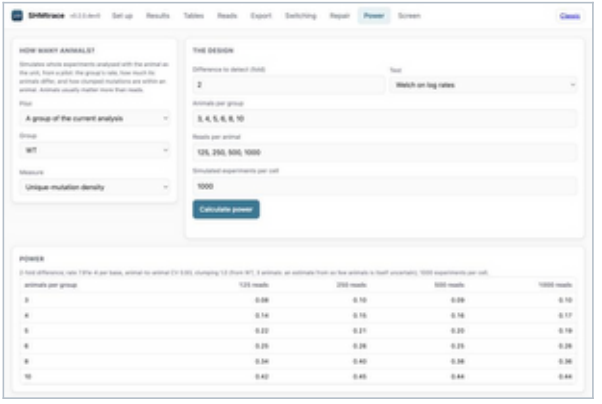

A
SHMtrace
Set up
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Classic
1 · REGION AND DATA TYPE
Region
jh4_intron
Mouse JH4 downstream intron, the neutral polyclonal assay.
Sequencing platform
snt
Calling mode
recommended
Quality-calibrated against your controls. The default.
Advanced
2 · REFERENCE
Built-in reference
JH4 – Base (1230 bp)
Drop the reference FASTA here, or click to choose
Reference: JH4 – Base
Contaminant FASTA (optional)
Optional: a sequence to exclude
Contaminant: Contaminant_JH4 (added for you)
3 · SAMPLES
Drop read files or a folder here, or click to choose
FASTQ, FASTA, BAM, Sanger .ab1 or GenBank .gb, plain or gzipped.
A dropped folder becomes one sample, and R1/R2 files are paired.
Choose a folder…
[private/tmp/claude-501/-Users-abdulmateenaderinto
Load sheet
6 samples from /private/tmp/claude-501/-Users-abdulmateenaderinto-Xenotrim/d53d3cae-8f63-4b59-8442-
Group by name
Controls by name
Clear
6 samples, 3 controls
WK2 1 file(s) from the sheet
WT
control
WK3 1 file(s) from the sheet
WT
control
WK4 1 file(s) from the sheet
WT
control
KO1 1 file(s) from the sheet
Akide-/-
control
KO2 1 file(s) from the sheet
Akide-/-
control
KO3 1 file(s) from the sheet
Akide-/-
control
A control is a sample with no real mutations (AID knockout, empty vector, germline). Controls measure the sequencing background for every other sample, and each control is measured against the others, never itself.
4 · RUN
JH4 intron, WT vs Akide-/-
Analyse


B
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JH4 INTRON, WT VS AICDA-/-
WT is clearly above Akide-/- · 65.8 vs 0.46 per 10⁶ bp · 137-fold · P = 0.0025
Region
JH4_intron · JH4 · window 101–629 of 1230 bp
Data
snt · mode recommended (qual) · 500 reads per sample
Samples
6 in 2 group(s), 3 control(s)
Unique mutations / 10⁶ bp
WT
Akide-/-
Mutation density:
unique mutations per 10⁶ bp; one dot per sample, bar is the geometric mean ± geometric SD
Mutations / 10⁶ bp
Mutation frequency
mutations per 10⁶ bases sequenced
WT (n = 618)
Akide-/- (n = 3)
WT
mutations


C
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Samples
Filter rows…
4 rows
Download CSV
sample
group
control
reads
unique mutations
unique density x1e5
one event density x1e5
mutation frequency x1e3
mutated reads pct
mean mutations per read
WK2
WT
500
205
77.69
0.079
2.703
21.4
1.43
WK3
WT
500
340
131.7
0.0972
22.45
91.8
11.88
WK4
WT
500
73
27.83
0.0972
4.934
36.6
2.61
KO1
Akide-/-
yes
500
0
0
0.0794
0
0
0
KO2
Akide-/-
yes
500
2
0.76
0.0796
0.1096
0.8
0.058
KO3
Akide-/-
yes
500
1
0.38
0.0802
0.0302
1.6
0.076


D
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HOW MANY ANIMALS?
Simulates whole experiments analysed with the animal as the unit, from a pilot: the group's rate, how much its animals differ, and how clumped mutations are within an animal. Animals usually matter more than reads.
Pilot
A group of the current analysis
Group
WT
Measure
Unique mutation density
THE DESIGN
Difference to detect (fold)
2
Test
Welch on log rates
Animals per group
3, 4, 5, 6, 8, 10
Reads per animal
125, 250, 500, 1000
Simulated experiments per cell
1000
Calculate power
POWER
2-fold difference, ratio 1.97e- 4 per base, animal-to-animal CV 0.93, clumping 1.0 (from WT). 3 animals: an estimate from so few animals is itself uncertain). 1000 experiments per cell.
animals per group
125 reads
250 reads
500 reads
1000 reads
3
0.08
0.10
0.09
0.10
4
0.14
0.15
0.16
0.17
5
0.22
0.21
0.20
0.19
6
0.25
0.26
0.25
0.26
8
0.34
0.40
0.36
0.36
10
0.42
0.45
0.44
0.44